

Weekly Lesson Plan – Week 8 · Semester 1 · 2026–27 · Da Vinci School

Teacher: Gavaldon Week of: September 28, 2026 Course: Principles of Applied Engineering				
TEKS / ELPS:				
TEKS: §127.391(d)(2)(C) computer-aided design and drafting (CADD) ELPS: c.1C, c.2C, c.3E, c.4C, c.4F, c.5G				
Aim of the Lesson / Objective: <i>Statement of what you want students to do (skill) and/or know (knowledge). Students will...</i>				
MONDAY SEP 28	TUESDAY SEP 29	WEDNESDAY SEP 30	THURSDAY OCT 1	FRIDAY OCT 2
Students will insert and calibrate a reference image and sketch a bottle profile using lines and a fit point spline.	Students will revolve a sketch profile into a solid and apply shell and fillet features to the resulting body.	Students will finish the bottle with an appearance and evaluate their own model against a checklist before submitting it.	Students will sketch a connected path and a profile and use the sweep feature to model a paperclip.	Students will complete the paperclip model and compare extrude, revolve and sweep in their own words.
Employed Language Domain(s) Objective: <i>How students achieve the objective – Listening, Speaking, Reading, Writing. Students will...</i>				
Students will collaborate in teams to investigate engineering problems, communicate their findings, and present their solutions. They will listen to team members' ideas, discuss potential solutions, read relevant technical materials, and write explanations of their designs and findings.				
Assessment of Content Objective: <i>Instrument used to determine mastery. Ex: Exit Ticket.</i>				
Students will be assessed by written, group, or oral responses throughout the class period.				
Set (Bell Work): <i>A motivating action that engages students in the objective – question, fact, activity. Essential Question:</i>				
MONDAY SEP 28	TUESDAY SEP 29	WEDNESDAY SEP 30	THURSDAY OCT 1	FRIDAY OCT 2
Look at a plastic water bottle. Why is the outside shaped with ridges and curves instead of being a smooth cylinder?	You find a mistake in a design your team already sent to the customer. Nobody else has noticed. Do you say something?	Why do bottles have a neck that is narrower than the body? Give two reasons.	A paperclip is a single bent wire and it has not changed in over 100 years. Why has nobody improved it?	You can make a product last twice as long for fifty cents more per unit, but then people buy half as many. What do you do?
Methodology / Steps: <i>What it takes to teach the objective so 100% can pass the assessment. Teacher will... Teams will... Individuals will...</i>				
MONDAY SEP 28	TUESDAY SEP 29	WEDNESDAY SEP 30	THURSDAY OCT 1	FRIDAY OCT 2
Step 1: Bell ringer / essential question. Step 2: Bell work: why is a plastic water bottle shaped with ridges instead of being a smooth cylinder? (10 min) Step 3: Save the bottle picture; new file saved in your folder (5 min) Step 4: Day 2 video: Insert > Canvas, calibrate, sketch half the outline (28 min) Step 5: Save (Ctrl+S) before the bell (2 min)	Step 1: Bell ringer / essential question. Step 2: Bell work (dilemma): you find a mistake in a design already sent to the customer - do you say something? (10 min) Step 3: Day 2 video: revolve around the center line, shell through the top, fillet the edges (30 min) Step 4: Save (Ctrl+S) (3 min) Step 5: Check: is it round from every angle? (2 min)	Step 1: Bell ringer / essential question. Step 2: Bell work: why is a bottle neck narrower than the body? Give two reasons. (10 min) Step 3: Finish the bottle: anything left, then drag on the glass appearance (27 min) Step 4: Self-check against the list; tell me if it is not going to be done (6 min) Step 5: Save (Ctrl+S) in Students > your name (2 min)	Step 1: Bell ringer / essential question. Step 2: Bell work: a paperclip has not changed in 100 years - why has nobody improved it? (10 min) Step 3: New file: Day 3 - Paperclip - Lastname, units in mm (5 min) Step 4: Day 3 video: sketch the path, sketch the circle, sweep (28 min) Step 5: Save (Ctrl+S) before the bell (2 min)	Step 1: Bell ringer / essential question. Step 2: Bell work (dilemma): make it last twice as long for 50 cents more, and sell half as many? (10 min) Step 3: Finish the sweep; adjust the wire thickness; save (22 min) Step 4: Folder check: block, bottle, paperclip - all three in Students > your name (8 min) Step 5: Three tools, three builds: write what each one does in your own words (5 min)
Resources: <i>Textbook, handouts, visuals, laptops.</i>				
Google Classroom, projector, engineering notebook, Design Brief packet / worksheets, laptops or phones for AI tools.				
Re-Teaching Activities: <i>What you prepare if students do not master the objective.</i>				
Review during bellwork; small-group re-teach at the teacher table; one-on-one explanation on request; peer and native-language support within the seating group.				
Reflection: <i>Students analyze what they accomplished and how they reached mastery.</i>				
Exit ticket – students state what they accomplished and what worked best.				
Study Skills: <i>Skills implemented by students – note-taking, summarizing, teamwork, etc.</i>				
Note-taking, summarizing, outlining, reading comprehension, teamwork, oral communication, problem-solving.				
SPED & 504 Accommodations / Modifications: <i>Codes.</i>				
Per Individualized Education Plan (IEP) / 504 plan.				
ELL Accommodations: <i>Codes.</i>				
Clarify directions (CD); Extra time (ET); Peer and Native Language Support (PN); Pre-teach vocabulary (PV); Rephrase, Repeat, Slow Down (RS); Visual Cues (VC); Wait time (WT); Pictorial Representation (DP). GROUPING: students are seated in mixed-proficiency small groups so emergent bilingual students hear and use the academic vocabulary in context, rehearse with a peer before whole-class response, and can be pulled as a small group for re-teach (PN, WT). ONE-ON-ONE: individual teacher explanation is available on request every period; directions are restated and modeled individually at the student desk after whole-class delivery (CD, RS, VC).				